

Date 30. Recommender systems

Making recommendations

Predicting movie ratings

User rates movies using one to five stars
zero

Movie	Alice(1)	Bob(2)	Carol(3)	Dave(4)
Love at last	5	5	0	0
Romance forever	5	?	?	0
Cute puppies of love	?	4	0	?
Nonstop car chases	0	0	5	4
Swords vs. karate	0	0	5	?

$n_u = 4$ $r(1,1) = 1$
 $n_m = 5$ $r(3,1) = 0$ $y^{(3,2)} = 4$

Ratings

★				
★	★			
★	★	★		
★	★	★	★	
★	★	★	★	★

n_u = no. of users
 n_m = no. of movies
 $r(i,j)=1$ if user j has rated movie i
 $y^{(i,j)}$ = rating given by user j to movie i
(defined only if $r(i,j)=1$)

Collaborative filtering

Using per-item features

What if we have features of the movies?

$n_u = 4$
 $n_m = 5$
 $n = 2$

Movie	Alice(1)	Bob(2)	Carol(3)	Dave(4)	x_1 (romance)	x_2 (action)
Love at last	5	5	0	0	0.9	0
Romance forever	5	?	?	0	1.0	0.01
→ Cute puppies of love	?	4	0	?	0.99	0
Nonstop car chases	0	0	5	4	0.1	1.0
Swords vs. karate	0	0	5	?	0	0.9

$$x^{(1)} = \begin{bmatrix} 0.9 \\ 0 \end{bmatrix}$$

$$x^{(3)} = \begin{bmatrix} 0.99 \\ 0 \end{bmatrix}$$

For user 1: Predict rating for movie i as: $w \cdot x^{(i)} + b$

← just linear regression

$$w^{(1)} = \begin{bmatrix} 5 \\ 0 \end{bmatrix} \quad b^{(1)} = 0 \quad x^{(3)} = \begin{bmatrix} 0.99 \\ 0 \end{bmatrix}$$

$$w^{(1)} \cdot x^{(3)} + b^{(1)} = 4.95$$

Cost function

Notation:

- $r(i,j) = 1$ if user j has rated movie i (0 otherwise)
- $y^{(i,j)}$ = rating given by user j on movie i (if defined)
- $w^{(j)}, b^{(j)}$ = parameters for user j
- $x^{(i)}$ = feature vector for movie i

For user j and movie i , predict rating: $w^{(j)} \cdot x^{(i)} + b^{(j)}$

- $m^{(j)}$ = no. of movies rated by user j
- To learn $w^{(j)}, b^{(j)}$

Cost function

Notation:

- $r(i,j) = 1$ if user j has rated movie i (0 otherwise)
- $y^{(i,j)}$ = rating given by user j on movie i (if defined)
- $w^{(j)}, b^{(j)}$ = parameters for user j
- $x^{(i)}$ = feature vector for movie i

For user j and movie i , predict rating: $w^{(j)} \cdot x^{(i)} + b^{(j)}$

- $m^{(j)}$ = no. of movies rated by user j
- To learn $w^{(j)}, b^{(j)}$

$$\frac{1}{2m^{(j)}} \sum_{i:r(i,j)=1} (w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)})^2$$

$$\min_{w^{(j)} b^{(j)}} J(w^{(j)}, b^{(j)}) = \frac{1}{2m^{(j)}} \left(\sum_{i:r(i,j)=1} (w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)})^2 \right) + \frac{\lambda}{2m^{(j)}} \sum_{k=1}^n (w_k^{(j)})^2$$

Cost function

To learn parameters $w^{(j)}, b^{(j)}$ for user j :

$$J(w^{(j)}, b^{(j)}) = \frac{1}{2} \sum_{i:r(i,j)=1} (w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)})^2 + \frac{\lambda}{2} \sum_{k=1}^n (w_k^{(j)})^2$$

To learn parameters $w^{(1)}, b^{(1)}, w^{(2)}, b^{(2)}, \dots, w^{(n_u)}, b^{(n_u)}$ for all users :

$$J\left(\begin{matrix} w^{(1)}, \dots, w^{(n_u)} \\ b^{(1)}, \dots, b^{(n_u)} \end{matrix}\right) = \frac{1}{2} \sum_{j=1}^{n_u} \sum_{i:r(i,j)=1} (w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)})^2 + \frac{\lambda}{2} \sum_{j=1}^{n_u} \sum_{k=1}^n (w_k^{(j)})^2$$

Collaborative filtering

Learn the features

Problem motivation

Movie	Alice (1)	Bob (2)	Carol (3)	Dave (4)	x_1 (romance)	x_2 (action)
Love at last	5	5	0	0	?	?
Romance forever	5	?	?	0	?	?
Cute puppies of love	?	4	0	?	?	?
Nonstop car chases	0	0	5	4	?	?
Swords vs. karate	0	0	5	?	?	?

$$w^{(1)} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}, w^{(2)} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}, w^{(3)} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}, w^{(4)} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

$$b^{(1)} = 0, b^{(2)} = 0, b^{(3)} = 0, b^{(4)} = 0 \leftarrow$$

using $w^{(j)} \cdot x^{(i)} + b^{(j)}$

$$\left. \begin{aligned} w^{(1)} \cdot x^{(1)} &\approx 5 \\ w^{(2)} \cdot x^{(1)} &\approx 5 \\ w^{(3)} \cdot x^{(1)} &\approx 0 \\ w^{(4)} \cdot x^{(1)} &\approx 0 \end{aligned} \right\} \rightarrow x^{(1)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Cost function

Given $w^{(1)}, b^{(1)}, w^{(2)}, b^{(2)}, \dots, w^{(n_u)}, b^{(n_u)}$

to learn $x^{(i)}$:

$$J(x^{(i)}) = \frac{1}{2} \sum_{j:r(i,j)=1} (w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)})^2 + \frac{\lambda}{2} \sum_{k=1}^n (x_k^{(i)})^2$$

→ To learn $x^{(1)}, x^{(2)}, \dots, x^{(n_m)}$:

$$J(x^{(1)}, x^{(2)}, \dots, x^{(n_m)}) = \frac{1}{2} \sum_{i=1}^{n_m} \sum_{j:r(i,j)=1} (w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)})^2 + \frac{\lambda}{2} \sum_{i=1}^{n_m} \sum_{k=1}^n (x_k^{(i)})^2$$

Collaborative filtering

	Alice	Bob	Carol
Movie1	5	5	?
Movie2	?	2	3

Cost function to learn $w^{(1)}, b^{(1)}, \dots, w^{(n_u)}, b^{(n_u)}$:

$$\min_{w^{(1)}, b^{(1)}, \dots, w^{(n_u)}, b^{(n_u)}} \frac{1}{2} \sum_{j=1}^{n_u} \sum_{i:r(i,j)=1} (w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)})^2 + \frac{\lambda}{2} \sum_{j=1}^{n_u} \sum_{k=1}^n (w_k^{(j)})^2$$

Cost function to learn $x^{(1)}, \dots, x^{(n_m)}$:

$$\min_{x^{(1)}, \dots, x^{(n_m)}} \frac{1}{2} \sum_{i=1}^{n_m} \sum_{j:r(i,j)=1} (w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)})^2 + \frac{\lambda}{2} \sum_{i=1}^{n_m} \sum_{k=1}^n (x_k^{(i)})^2$$

Put them together:

$$\min_{\substack{w^{(1)}, \dots, w^{(n_u)} \\ b^{(1)}, \dots, b^{(n_u)} \\ x^{(1)}, \dots, x^{(n_m)}}} J(w, b, x) = \frac{1}{2} \sum_{(i,j):r(i,j)=1} (w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)})^2 + \frac{\lambda}{2} \sum_{j=1}^{n_u} \sum_{k=1}^n (w_k^{(j)})^2 + \frac{\lambda}{2} \sum_{i=1}^{n_m} \sum_{k=1}^n (x_k^{(i)})^2$$

Gradient descent

Gradient Descent

collaborative filtering

Linear regression (course 1)

repeat {

~~$$w_i = w_i - \alpha \frac{\partial}{\partial w_i} J(w, b)$$~~

~~$$b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$~~

$$w_i^{(j)} = w_i^{(j)} - \alpha \frac{\partial}{\partial w_i^{(j)}} J(w, b, x)$$

$$b^{(j)} = b^{(j)} - \alpha \frac{\partial}{\partial b^{(j)}} J(w, b, x)$$

$$x_k^{(i)} = x_k^{(i)} - \alpha \frac{\partial}{\partial x_k^{(i)}} J(w, b, x)$$

}

parameters w, b, x

x is also a parameter

Binary labels: favs, likes and clicks

Binary labels

Movie	Alice(1)	Bob(2)	Carol(3)	Dave(4)
Love at last	1	1	0	0
Romance forever	1	? ←	? ←	0
Cute puppies of love	? ←	1	0	? ←
Nonstop car chases	0	0	1	1
Swords vs. karate	0	0	1	? ←

Example applications

- 1. Did user j purchase an item after being shown? 1, 0, ?
- 2. Did user j fav/like an item? 1, 0, ?
- 3. Did user j spend at least 30sec with an item?
- 4. Did user j click on an item?

Meaning of ratings:

1 - engaged after being shown item

0 - did not engage after being shown item

? - item not yet shown

From regression to binary classification

- Previously:
- Predict $y^{(i,j)}$ as $\underline{w^{(j)} \cdot x^{(i)} + b^{(j)}}$
- For binary labels:
Predict that the probability of $y^{(i,j)} = 1$
is given by $g(w^{(j)} \cdot x^{(i)} + b^{(j)})$

$$\text{where } \underline{g(z) = \frac{1}{1+e^{-z}}}$$

Cost function for binary application

Previous cost function:

$$\frac{1}{2} \sum_{(i,j): r(i,j)=1} (\underbrace{w^{(j)} \cdot x^{(i)} + b^{(j)}}_{f(x)}) - y^{(i,j)})^2 + \frac{\lambda}{2} \sum_{i=1}^{n_m} \sum_{k=1}^n (x_k^{(i)})^2 + \frac{\lambda}{2} \sum_{j=1}^{n_u} \sum_{k=1}^n (w_k^{(j)})^2$$

Loss for binary labels $y^{(i,j)}$: $f_{(w,b,x)}(x) = g(w^{(j)} \cdot x^{(i)} + b^{(j)})$

$$L(f_{(w,b,x)}(x), y^{(i,j)}) = -y^{(i,j)} \log(f_{(w,b,x)}(x)) - (1 - y^{(i,j)}) \log(1 - f_{(w,b,x)}(x))$$

Loss for
single
example

$$J(w, b, x) = \sum_{(i,j): r(i,j)=1} L(f_{(w,b,x)}(x), y^{(i,j)})$$